

Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in $M \geq 6.5$ Earthquake Catalogs (2150 BCE – 2026 CE)

© 2026 Yaroslav Marshalkin

Yaroslav Marshalkin

Corresponding author: marshalkin@gmail.com

Telegram: @MRSHLKN | GitHub: github.com/marshalkin-ux/paleoseismic-clustering

Abstract.

We analyse a merged catalog of 4,267 unique $M \geq 6.5$ events (from 4,418 CSV rows; ~151 NOAA $M < 6.5$ excluded from clustering; 2150 BCE–2026 CE) using the Baiesi–Paczuski metric η with tectonic-path distance (Bird 2003). 47 global seismic series are identified (27 modern, 15 early, 5 historical candidates). Significance: permutation test ($n=10,000$, $p < 0.0001$, $z = -6.17$); ETAS validation ($FPR=0/100$, $seed=42$); FDR (45/47 at $q=0.05$). Series detection is a statistical anomaly, not proof of causal triggering. Largest series: 1905–1910 (193 events, 43 regions); most extensive modern: S170 (46 events, 12 regions, $M_{max}=9.1$).

Keywords: global seismicity; seismic series; earthquake clustering; tectonic distance; Baiesi–Paczuski metric; ETAS validation; false discovery rate; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake (M_w 7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra–Andaman earthquake (M_w 9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that $M \geq 7$ clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global $M \geq 7$ and $M \geq 8$ rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi–Paczuski [2004] metric and Zaliapin–Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. We test the hypothesis that multi-regional seismic series exist in a four-millennium catalog of $M \geq 6.5$ earthquakes, using an adapted eta metric with tectonic distance along Bird (2003) plate boundaries. **Novelty.** To our knowledge, this is the first global application of nearest-neighbor clustering with tectonic-path distance across historical, early instrumental, and modern catalogs, combined with ETAS validation and FDR correction.

2. DATA AND METHODS

2.1 Catalog compilation

Three catalogs were merged: USGS ComCat (1900–2026, primary instrumental); ISC Bulletin (relocated hypocenters); and NOAA NGDC (historical and paleoseismic records from ~2150 BCE). Duplicates were merged using ± 30 days and ≤ 50 km tolerance; source priority: ISC > USGS > NOAA. After deduplication, the catalog contains 4,267 unique $M \geq 6.5$ events (from 4,418 CSV rows; ~151 $M < 6.5$ excluded from clustering). USGS ComCat: 2,088 raw (1973–2026) $\rightarrow$ 2,041 after merge/ISC. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041). `quality_score` is metadata, not an inclusion filter. Final catalog: 4,267 $M \geq 6.5$ events (4,418 CSV rows).

2.1.1 Catalog completeness

Maximum-curvature analysis yields $M_c = 6.55$. Maximum-likelihood b-value from 1,688 events above M_c :

$$b = 0.911 \pm 0.018 \quad (n = 1,688 \text{ events})$$

2.2 Tectonic distance and connectivity metric

Tectonic distance r_{ij} is the shortest path between hypocenters along the Bird (2003) plate-boundary graph (20 key segments). Paths are computed with Dijkstra's algorithm (NetworkX).

If either hypocenter lies >500 km from the nearest boundary node, or no graph path exists, $r_{ij} = 1.5 \times r_{GC}$ (great-circle Haversine). Limitations: 500 km snap and $1.5 \times GC$ are approximations; intraplate pairs rely on GC penalty; sensitivity deferred to future work. Qualitatively improves inter-plate eta links by $\sim 0.3 \log_{10}$.

$$\text{eta}_{ij} = t_{ij} * r_{ij}^{1.6} * 10^{(-b \cdot m_i)}$$

b=1.0 per Baiesi & Paczuski (2004) for eta comparability; catalog b=0.911+/-0.018 in Monte Carlo null only.

2.3 Analysis pipeline

Raw catalogs → dedup → exclude $M < 6.5$ (~151 NOAA) → GK declustering (primary) → eta NN forest → sliding windows → merge overlapping → series criteria → MC/ETAS/FDR. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041) — separate sensitivity check, not sequential filter.

2.4 Threshold eta₀ and series detection

Automatic eta₀ from nearest-neighbor eta distribution: KDE valley between bimodal $\log_{10}(\text{eta})$ modes (Zaliapin & Ben-Zion 2013); default eta₀ = $10^{\text{median } \log_{10} \text{ eta}}$. Validated against Poisson permutation null (n = 10,000).

1. Declustering via Gardner–Knopoff [1974].
2. Nearest-neighbor forest: parent $i^* = \text{argmin } \text{eta}_{ij}$ subject to $\text{eta}_{ij} < \text{eta}_0$.
3. Global series: $N \geq 4$; $M \geq 6.5$; ≥ 3 Flinn–Engdahl regions.
4. Sliding windows (1, 2, 5 yr); overlapping groups merged.

2.5 Statistical validation

Permutation test: n = 10,000, $p < 0.0001$, $z = -6.17$ (modern). ETAS validation: $\mu=0.008$, $K=0.08$, $\alpha=1.0$, $c=0.005$, $d, p=1.1$; 500 km cutoff; 100 catalogs (seed=42). FPR=0/100; $p_{\text{ETAS}}=0.0000$. Limitation (see Discussion): FPR=0/100 at seed=42 only; multi-seed robustness recommended. FDR ($q=0.05$): 45/47 significant. Declustering: GK 2,017/2,041; ZBZ 2,040/2,041.

3. RESULTS

3.1 Identified series

Full historical analysis yields 47 global seismic series: 27 modern ($p < 0.0001$), 15 early instrumental ($p = 0.007$; pre-1960 incompleteness caveat), 5 historical candidates (not significant, $p = 0.46$; 47 $M \geq 6.5$ events pre-1900). 142 cluster candidates before filtering.

ID	N	Regions	M _{max}	Period	Notes
1905–1910	193	43	8.8	1905–1910	Largest series; early instrumental
S047	53	5	8.0	1982–2024	Western Pacific subduction corridor
S170	46	12	9.1	2002–2023	Sunda belt; Sumatra 2004 (M 9.1)
S095	25	4	7.9	1989–2017	Western Pacific arc

S116	22	5	8.2	1993–2021	South Pacific multi-arc series
------	----	---	-----	-----------	--------------------------------

Table 1. Top five multi-regional series (≥ 3 Flinn–Engdahl regions).

3.2 Spatial–temporal distribution

Elevated series activity occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic distance lowers the η_0 threshold by $\sim 0.3 \log_{10}$ units compared with Euclidean distance.

4. DISCUSSION AND CONCLUSIONS

Statistical anomaly (established): series incompatible with Poisson and local-only ETAS nulls; FDR 45/47. Conclusion about η links and p-values, not causality. Working hypotheses (not claims): viscoelastic mantle coupling (Pollitz 1998), dynamic triggering (Hill 1993; Brodsky 2006), shared tectonic loading (Freed & Lin 2001). ETAS seed limitation: FPR=0/100 at seed=42 only; multiseed future work. Limitations: tectonic distance approximations.

1. 4,267 $M \geq 6.5$ events (4,418 CSV records) contain 47 global series; 27 modern significant at $p < 0.0001$.
2. ETAS validation ($\mu=0.008$, $K=0.08$, $\alpha=1.0$, $c=0.005$ d, $p=1.1$, 500 km; FPR=0/100) and FDR (45/47) confirm non-randomness.
3. Largest series: 1905–1910 (193 events, 43 regions, $M_{\max}=8.8$); most spatially extensive modern: S170 (46 events, 12 regions, 2002–2023, $M_{\max}=9.1$).
4. Tectonic distance increases η sensitivity by $\sim 0.3 \log_{10}$.
5. Interpretive fork: (a) if η links are real — hazard implications without claiming direct triggering; (b) if artifacts — FDR+ETAS remains reproducible null-test. Co-occurrence may reflect shared tectonic loading.

Future work: DeltaCFS for S047, S170, S095; multi-seed ETAS; Zenodo.

DATA AND CODE AVAILABILITY

Data and code: github.com/marshalkin-ux/paleoseismic-clustering (docs/data_availability.md).

REFERENCES

1. Baiesi M., Paczuski M. (2004). Scale-free networks of earthquakes and aftershocks. *Phys. Rev. E*, 69, 066106.
2. Zaliapin I. et al. (2008). Clustering analysis of seismicity and aftershock identification. *Phys. Rev. Lett.*, 101, 018501.
3. Zaliapin I., Ben-Zion Y. (2013). Earthquake clusters in southern California I. *J. Geophys. Res.*, 118, 2847–2864.
4. Bird P. (2003). An updated digital model of plate boundaries. *Geochem. Geophys. Geosyst.*, 4(3), 1027.
5. Hill D.P. et al. (1993). Seismicity remotely triggered by the magnitude 7.3 Landers earthquake. *Science*, 260, 1617–1623.

6. Brodsky E.E., Prejean S.G. (2006). New constraints on mechanisms of remotely triggered seismicity. *J. Geophys. Res.*, 111, B06312.
7. Pollitz F.F. et al. (1998). Viscosity of oceanic asthenosphere inferred from remote triggering. *Science*, 280, 1245–1249.
8. King G.C.P. et al. (1994). Static stress changes and the triggering of earthquakes. *BSSA*, 84, 935–953.
9. Stein R.S. (1999). The role of stress transfer in earthquake occurrence. *Nature*, 402, 605–609.
10. Michael A.J. (2011). Random variability explains apparent global clustering of large earthquakes. *Geophys. Res. Lett.*, 38, L21301.
11. Shearer P.M., Stark P.B. (2012). Global risk of big earthquakes has not recently increased. *PNAS*, 109(3), 717–721.
12. Ogata Y. (1988). Statistical models for earthquake occurrences and residual analysis. *J. Amer. Stat. Assoc.*, 83, 9–27.
13. Benjamini Y., Hochberg Y. (1995). Controlling the false discovery rate. *J. Roy. Stat. Soc. B*, 57(1), 289–300.
14. Gardner J.K., Knopoff L. (1974). Is the sequence of earthquakes in Southern California Poissonian? *BSSA*, 64, 1363–1367.
15. Woessner J., Wiemer S. (2005). Assessing the quality of earthquake catalogues. *BSSA*, 95, 684–698.
16. Young J.B. et al. (1996). The Flinn–Engdahl regionalization scheme: the 1995 revision. *Phys. Earth Planet. Int.*, 96, 223–297.

Acknowledgments. The author acknowledges USGS, NOAA NGDC, and the ISC for maintaining open seismic catalogs. This work received no targeted funding.